Most Common SQL Window Functions

These are the ones you'll see constantly in analytics, reporting, Databricks, SQL Server, Snowflake, etc.

Ranking Functions

```
ROW_NUMBER()
```

Assigns a unique sequence.

```
ROW_NUMBER() OVER (
    PARTITION BY Customer
    ORDER BY Sales DESC
)
```

Use for:

- Top 1 row per group
- Latest record per customer
- De-duplication

```
RANK()
```

Ties get the same rank, with gaps.

Sales	Rank
100	1
90	2
90	2
80	4

```
DENSE_RANK()
```

Ties get the same rank, no gaps.

Sales	Dense Rank
100	1
90	2
90	2
80	3

⬅➡ Previous / Next Row

```
LAG()
```

Previous row value.

```
LAG(Sales) OVER (
    PARTITION BY Product
    ORDER BY Month
)
```

Used for:

- Month-over-month change
- Prior inventory
- Prior pricing

```
LEAD()
```

Next row value.

```
LEAD(Sales) OVER (
    PARTITION BY Product
    ORDER BY Month
)
```

Used for:

- Forecasting
- Looking ahead
- Gap analysis

Running Totals

```
SUM()
```

```
SUM(Sales) OVER (
    PARTITION BY Customer
    ORDER BY OrderDate
)
```

Produces:

Date	Sales	Running Total
Jan	100	100
Feb	200	300
Mar	150	450

Very common in BI.

Moving Averages

```
AVG()
```

```
AVG(Sales) OVER (
    ORDER BY Month
    ROWS BETWEEN 2 PRECEDING AND CURRENT ROW
)
```

3-month moving average.

First / Last Value

```
FIRST_VALUE()
LAST_VALUE()
```

Example:

```
FIRST_VALUE(Price) OVER(
    PARTITION BY Product
    ORDER BY EffectiveDate
)
```

Useful for:

- Original price
- First status
- First order

Aggregate Without Collapsing Rows

One of the coolest window function uses.

```
SUM(Sales) OVER(PARTITION BY Customer)
```

Returns customer total on every row:

Customer	Order	Sales	Customer Total
A	1	100	450
A	2	200	450
A	3	150	450

Unlike a `GROUP BY`, you keep detail rows.

If I had to learn only 5

For BI/Data Engineering/Data Analytics:

1. `ROW_NUMBER()`
2. `LAG()`
3. `LEAD()`
4. `SUM() OVER()`
5. `AVG() OVER()`

Those five probably solve **80-90%** of real-world window function problems you'll encounter in SQL Server, Databricks SQL, Snowflake, or BigQuery.

> For your Tableau/KNIME/Product reporting work, `ROW_NUMBER()` is by far the one you'll use most often for "latest record per product/customer/MLINE" situations.